

Final Exam: Principles of Fixed Income

Cello Research

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Due: Monday, June 29th by 9:00am.

Problem 1. (10 points) You are considering the purchase of a home. It is in excellent condition, except for the roof. The roof has only 5 years of life remaining. A new roof lasts 20 years but costs \$20,000. The house is expected to last forever. Assuming that costs will remain constant and the annual interest rate is 5% what value would you assign to the existing roof? [Hint: Consider the tradeoff in costs between replacing the roof now or waiting 5 years. Take into account the fact that a roof must be replaced every 20 years in your calculations.]

Solution. The value of the existing roof is the difference between the PV of replacing the roof now or waiting 5 years. The PV of the cash flows associated with replacing the roof now are:

$$\begin{aligned}\text{PV-Replace} &= 20000 + \frac{20000}{(1+5\%)^{20}} + \frac{20000}{(1+5\%)^{40}} + \cdots \\ &= \frac{20000}{1 - \frac{1}{(1+5\%)^{20}}} \\ &= 32,097.03\end{aligned}$$

The PV of the cash flows associated with waiting 5 years are:

$$\begin{aligned}\text{PV-Wait} &= \frac{20000}{(1+5\%)^5} + \frac{20000}{(1+5\%)^{25}} + \cdots \\ &= \frac{\frac{20000}{(1+5\%)^5}}{1 - \frac{1}{(1+5\%)^{20}}} \\ &= 25,148.86\end{aligned}$$

The difference between the PV of the two cash flows is the current value of the roof: \$6,948.17.

Problem 2. (10 points) Consider the two projects whose cash flows are shown in the table below. Find the IRRs of the two projects and the NPVs at an annual interest rate of 5%. Show that the IRR and NPV figures yield different recommendations. Which one would you rather pick? Justify your choice.

Solution. The IRR of Project 1 is 15.24% and that of Project 2 is 12.38%. The NPV of Project 1 is \$29.88 and that of Project 2 is \$31.84. Clearly, the NPV analysis suggests Project 2, while the IRR analysis suggests Project 1.

To see which one we prefer, imagine two choices. First, you can borrow 150 and invest it in Project 2 for a NPV of \$31.84. Second, you can borrow 150, invest 100 of it in Project 1 for a NPV of \$29.88 and still have \$50 dollars in cash at the end of it all. The NPV of this choice is \$79.88. Thus, you should pick Project 1. The takeaway is that when you are considering two projects with different amounts of initial cash investment, you have to take into account the opportunity cost of the “extra cash” that corresponds to the difference in the initial investment amounts.

Years	Project 1 Cash Flow	Project 2 Cash Flow
0	-100	-150
1	30	42
2	30	42
3	30	42
4	30	42
5	30	42

Table 1: Cash flows for two projects

Problem 3. (10 points) An investor purchased a small apartment building for \$250,000. She made a downpayment of \$50,000 and financed the remaining balance with a 30-year fixed-rate mortgage at 12% annual interest, compounded monthly. For exactly 20 years she made equal-sized monthly payments as required by the terms of the loan. Now she has the opportunity to restructure the mortgage by refinancing the balance. She could borrow the current balance, pay off the original loan, and assume a new loan for this balance (Assume that no points or any charges are involved in the transaction). The new loan is a 20-year fixed-rate loan at 9% annual interest, compounded monthly, to be paid in equal monthly installments. Suppose that she receives an annual rate of 5% on her risk-free savings account. Should she restructure the mortgage?

Solution. The initial amount of the loan is \$200,000 and the fixed monthly payment is \$2,057.23. The balance of the loan after 20 years is given by the formula for the outstanding balance of a level-payment mortgage after k periods ($k = 240$ in this case): \$143,389.67. The new monthly payment on the 20-year fixed-rate loan at 9% is \$1,290.11. To decide whether the investor should refinance, we calculate the NPV @5% of the two cash flow streams: 10 years of monthly payments of \$2,057.23 versus 20 years of monthly payments of \$1,290.11. The NPV @5% of the remaining payments on the existing mortgage is \$193,957.97. The NPV @5% of the payments on the new mortgage is \$195,484.94. Therefore, she should not restructure the mortgage.

Problem 4. (15 points) Assume a continuous compounding convention for this problem.

1. What is the (modified) duration of a 20-year zero-coupon bond?
2. What is the convexity of a 20-year zero-coupon bond?
3. A portfolio manager tells you: "Convexity is the change in duration as a function of interest rates and since the duration of a zero-coupon bond is a constant that is independent of the interest rate, it follows that the convexity of a zero-coupon bond must be zero." Does this statement agree with your calculation above? If not, can you explain the flaw in the portfolio manager's reasoning?

Solution. (1) Assume that the zero-coupon bond has a face of \$100. Then, under a continuous compounding convention, its price P is given by $P = e^{-y \cdot 20}$, where y is the yield of the bond. The modified duration D is:

$$\begin{aligned}
 D &= -\frac{1}{P} \frac{dP}{dy} \\
 &= -e^{y \cdot 20} \times (-20)e^{-y \cdot 20} \\
 &= 20
 \end{aligned}$$

(2) The convexity C of a bond is given by $C = \frac{1}{P} \frac{d^2 P}{dy^2}$. For the zero-coupon bond, the second derivative is given by:

$$\begin{aligned}
 \frac{dP}{dy} &= (-20)100e^{-y \cdot 20} \\
 \frac{d^2 P}{dy^2} &= (20^2)100e^{-y \cdot 20}
 \end{aligned}$$

Plugging into the formula for C gives us a convexity of 400.

(3) The portfolio manager is incorrect in that they are using a sloppy definition of convexity – while duration does contain a first derivative of P and convexity does contain a second derivative, it is not true that $dD/dy = C$ as you can verify for yourself by differentiating the formula for D with respect to y . However, if we consider **dollar duration** dP/dy , then it is true that **dollar convexity** is the derivative of dollar duration and hence measures the rate of change of dollar duration as a function of interest rates.

Problem 5. (15 points) Portfolio X is only invested in zero-coupon bonds of a fixed maturity. It lost \$1.6 billion out of a market value of \$7.5 billion in assets because of an unexpected increase in interest rates from 3% to 5.7%. Assume all interest rates are continuously compounded.

1. What was the maturity of the zero-coupon bonds in Portfolio X?
2. By borrowing money, a new portfolio (Portfolio Y) of \$7.5 billion in zero-coupon bonds of a fixed maturity was levered up to \$20.5 billion. Assume that the liability was taken on through repo positions which are financed at the overnight rate. What is the total amount of the liability (L) necessary to lever up the portfolio and what is its duration (D_L)?
3. What maturity do the zero-coupon bonds in Portfolio Y have to have in order to generate the same losses as Portfolio X?
4. **(Extra Credit: 10 points)** A portfolio with both assets (A) and liabilities (L) has an equity (E) given by $E = A - L$. By treating equity as a portfolio, calculate the duration of equity D_E as a function of duration of the assets D_A and liabilities D_L . Use this formula to calculate the duration of the levered portfolio (Portfolio Y). What conclusions about the riskiness of Portfolio X versus Portfolio Y can you reach by going through this exercise?

Solution. (1) Under a continuous compounding convention, we know that the duration of a zero-coupon is equal to its maturity T . In equations:

$$\begin{aligned} T &= -\frac{1}{P} \frac{dP}{dy} \\ &= -\frac{1}{7.5} \frac{(-1.6)}{2.7\%} \\ &= 7.90 \text{ years} \end{aligned}$$

(2) The assets of the portfolio increased from \$7.5 billion to \$20.5 billion, so portfolio Y must have borrowed \$13.0 billion. Borrowing money overnight clearly has next to zero duration so we can assume $D_L \approx 0$.

(3) Applying the same reasoning as in (1) above, the duration of the zero-coupon bonds in Portfolio Y is equal to their maturity:

$$\begin{aligned} T &= -\frac{1}{P} \frac{dP}{dy} \\ &= -\frac{1}{20.5} \frac{(-1.6)}{2.7\%} \\ &= 2.89 \text{ years} \end{aligned}$$

(4) Since $E = A - L$ and we can regard E as a portfolio, we use the fact that the duration of a portfolio is equal to the weighted average of the durations of the individual components, where the weights are the relative price contribution of each component to the total portfolio:

$$D_E = \frac{A}{A - L} D_A + \frac{-L}{A - L} D_L$$

We have $D_L \approx 0$, $D_A = 2.89$ years, and $A/A - L = 20.5/7.5 = 2.73$. Plugging into the equation for D_E we obtain $D_E = 7.89$ years. Thus, the duration of the levered portfolio Y is the same as the initial portfolio X

and viewed from this lens both portfolios have the same risk. However, notice how leverage amplifies risks – by taking a portfolio of zero-coupon bonds that have a duration of 2.89 and leveraging them up we can end up with the same risks as owning an unlevered portfolio with a much greater duration.

Problem 6. (15 points) The principles of cash flow analysis can be used to evaluate the value of publicly traded corporations. The owner of a share of stock in a company can be expected to receive periodic dividends. Suppose that it is known that in year k , $k = 1, 2, \dots$, a dividend of D_k will be received. If the interest rate is fixed at r , it is reasonable to assign a value of the firm to the stockholders at the present value of this infinite dividend stream; namely:

$$V_0 = \frac{D_1}{1+r} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} + \dots$$

Note that the formula requires the future dividends be known. A popular way to specify dividends is to use the **constant-dividend model**, where dividends grow at a constant rate g . In particular, we assume that $D_{k+1} = (1+g)D_k$.

1. Under what assumptions does the infinite series expression for V_0 have a finite value?
2. Assuming that the summation has finite value, find a formula for V_0 .
3. The XX Corporation has just paid a dividend of \$1.37mm. The company is expected to grow at 10% for the foreseeable future, and hence most analysts project a similar growth in dividends. The discount rate used for this type of company is 15%. What is the value of a share of stock in the XX Corporation? (Assume that there are 1 million shares outstanding.)

Solution. (1) We have that:

$$\begin{aligned} V_0 &= \frac{D_1}{1+r} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} + \dots \\ &= D_1 \sum_{k=1}^{\infty} \frac{(1+g)^{k-1}}{(1+r)^k} \end{aligned}$$

The infinite sum is an infinite geometric series with ratio term $\frac{1+g}{1+r}$. This geometric series converges when $|\frac{1+g}{1+r}| < 1$. Assuming that $g, r > 0$, we see that this happens when $g < r$.

(2) If $g < r$, then $\sum_{k=1}^{\infty} \frac{(1+g)^{k-1}}{(1+r)^k}$ is a convergent geometric series with first term $1/(1+r)$ and ratio term $\frac{1+g}{1+r}$. Using the formula for the sum of an infinite geometric series and the expression for V_0 we obtain:

$$V_0 = \frac{D_1}{r-g}$$

(3) We know that $D_1 = \$1.37\text{mm}$, $g = 10\%$, and $r = 15\%$. Plugging in to the expression for V_0 gives us \$30,140,000. Since there are 1 million shares outstanding, each share is worth \$30.14.

Problem 7. (45 points) Often a mortgage payment is divided into a principal payment stream and an interest payment stream, and the two streams are sold separately. We will calculate the values of these individual streams in this problem. Consider a standard fixed-rate level-payment mortgage with an initial balance $B = B(0)$ with equal periodic payments of an amount M . Assume that the interest rate per period is r .

1. Show that the mortgage balance $B(k)$ after the k th payment satisfies:

$$B(k) = (1+r)B(k-1) - M$$

for $k = 0, 1, \dots$.

2. Show that the equation above has the solution:

$$B(k) = (1+r)^k B - \left[\frac{(1+r)^k - 1}{r} \right] M$$

3. Suppose that the mortgage has N periods and M is chosen so that $B(N) = 0$; show that:

$$M = \frac{r(1+r)^N B}{(1+r)^N - 1}$$

4. The k th payment has a principal component of $P(k) = M - rB(k-1)$. Show that $P(k) = (1+r)^{k-1}(M - rB)$.

5. Find the present value V (at rate r) of the principal payment stream in terms of B, r, N, M .

6. Find V in terms of r, N, M only.

7. What is the present value W of the interest payment stream $I(k) = rB(k-1)$ in terms of B and V ?

8. What is the value of V as $N \rightarrow \infty$?

9. For a given N , which stream do you think has the larger duration – principal or interest?

Solution. (1) In each time period k , the fixed periodic payment M consists of principal $P(k)$ and interest $I(k)$. We also have that $I(k) = rB(k-1)$ (in the problem r is the per-period interest rate). Then:

$$\begin{aligned} B(k) &= B(k-1) - P(k) \\ &= B(k-1) - (M - I(k)) \\ &= (1+r)B(k-1) - M \end{aligned}$$

(2) Use the recursive formula to see what the pattern for the $B(k)$'s is:

$$B(1) = (1+r)B - M$$

$$\begin{aligned} B(2) &= (1+r)B(1) - M \\ &= (1+r)[(1+r)B - M] - M \\ &= (1+r)^2 B - M[1 + (1+r)] \end{aligned}$$

$$\begin{aligned} B(3) &= (1+r)B(2) - M \\ &= (1+r)\{(1+r)^2 B - M[1 + (1+r)]\} - M \\ &= (1+r)^3 B - M[1 + (1+r) + (1+r)^2] \end{aligned}$$

It's clear that the general pattern is:

$$B(k) = (1+r)^k B - M[1 + (1+r) + (1+r)^2 + \cdots + (1+r)^{k-1}]$$

We obtain the desired expression noting that the expression in square brackets is a finite geometric series with initial term 1 and a ratio term of $1+r$.

(3) Plug $k = N$ into the expression for $B(k)$ and solve for M after noting that by hypothesis $B(N) = 0$.

(4) Plug in the expression for $B(k-1)$ into the formula for $P(k)$ and then simplify.

(5) The present value of the principal payments is:

$$\begin{aligned} V &= \sum_{k=1}^N \frac{P(k)}{(1+r)^k} \\ &= \sum_{k=1}^N \frac{(1+r)^{k-1}(M-rB)}{(1+r)^k} \\ &= \frac{N(M-rB)}{1+r} \end{aligned}$$

(6) Use the expression for M in terms of B to solve for B and then plug into the equation for V .

$$V = \frac{N}{1+r} \left[\frac{rM}{(1+r)^N - 1} \right]$$

(7) We know $M = P(k) + I(k)$. Therefore:

$$\begin{aligned} \text{PV of monthly payments} &= \text{PV principal cash flow stream} + \text{PV interest cash flow stream} \\ &= V + W \end{aligned}$$

However, the present value of the monthly mortgage payments is just the initial balance of the mortgage. We conclude that $W = B - V$.

(8) As $N \rightarrow \infty$ we have $V \rightarrow 0$ since $(1+r)^N$ grows a lot more quickly than N . (For a more rigorous proof, apply L'Hopital's Rule.)

(9) Clearly $I(k) = rB(k-1)$ is a decreasing function of k since the balance of an amortizing mortgage is reduced by the principal component of the fixed monthly payment over time. Since $M = P(k) + I(k)$, we conclude that $P(k)$ is an increasing function of k . Since duration is a weighted sum of the times k , with the weights being proportional to the cash flows at those times. Hence, the duration of the cash flow stream determined by $P(k)$ should be greater since more relative weight is given to higher values of k .

Problem 8. (25 points) Assume that all interest rates are compounded annually for this question. Suppose that the prices of \$1,000 face zero-coupon bonds are as follows:

Maturity	Price
1 year	\$950
2 year	\$890
3 year	\$830
4 year	\$770
5 year	\$720

Table 2: Zero coupon prices

1. Calculate the value of an instrument paying \$1,500 at the end of Year 2 and \$2,000 at the end of Year 5 without computing spot rates.
2. Compute the spot rates of interest associated with these zero-coupon prices.
3. Compute the implicit forward rates $f_t = f_{t,t+1}$, where $f_{t,t+1}$ is the forward rate that applies between time t and $t+1$ for $t = 0, 1, 2, 3, 4$ years.
4. By using the general formula that connects the forward rate $f_{i,j}$ to the spot rates s_i and s_j , show that the t -year spot rate s_t is given by the formula for $t \geq 2$:

$$s_t = \sqrt[t]{(1+f_0)(1+f_1)\cdots(1+f_{t-1})} - 1$$

This formula shows that spot rates can be considered geometric averages of the forward rates of interest.

5. Using the forward rates $f_{t,t+1}$ computed previously, calculate the price of a zero-coupon bond that pays \$10,000 at the end of Year 4. Does the computed value match the price shown in the table? Why or why not?

(1) Instead of using spot rates, we can use discount factors. Recall that the discount factor for time t is given by:

$$d_t = \frac{1}{(1 + s_t)^t}$$

In particular, we can read discount factors directly from the prices of zero-coupon bonds. For example, the price of the 2-year zero-coupon bond is given by:

$$\begin{aligned} \$890 &= \frac{\$1,000}{(1 + s_2)^2} \\ \implies \frac{\$890}{\$1,000} &= \frac{1}{(1 + s_2)^2} \\ \implies \frac{\$890}{\$1,000} &= d_2 \\ \implies d_2 &= 0.89 \end{aligned}$$

Similarly, $d_5 = 0.72$. The price of the instrument is therefore:

$$\begin{aligned} P &= d_2 \cdot \$1,500 + d_5 \cdot \$2,000 \\ &= 0.89 \cdot \$1,500 + 0.72 \cdot \$2,000 \\ &= \$2,775 \end{aligned}$$

(2) We have $s_1 = 5.26\%$, $s_2 = 6.00\%$, $s_3 = 6.41\%$, $s_4 = 6.75\%$, $s_5 = 6.79\%$.

(3) The equation for these forward rates is given by (for $1 \leq t \leq 4$):

$$\begin{aligned} (1 + s_t)^t (1 + f_{t,t+1})^1 &= (1 + s_{t+1})^{t+1} \\ \implies f_{t,t+1} &= \frac{(1 + s_{t+1})^{t+1}}{(1 + s_t)^t} - 1 \end{aligned}$$

When $t = 0$, we have that $f_{0,1} = s_1 = 5.26\%$. The remaining values can be calculated through the formula to obtain $f_{1,2} = 6.75\%$, $f_{2,3} = 7.23\%$, $f_{3,4} = 7.78\%$, and $f_{4,5} = 6.95\%$. (4) We already know that $s_1 = f_{0,1} = f_0$. Now, let's obtain an expression for s_2 :

$$\begin{aligned} (1 + s_1)^1 (1 + f_{1,2})^1 &= (1 + s_2)^2 \\ \implies (1 + f_0)(1 + f_1) &= (1 + s_2)^2 \\ \implies s_2 &= \sqrt{(1 + f_0)(1 + f_1)} - 1 \end{aligned}$$

The idea is clear – at each step we can bootstrap the expression for s_t in terms of f_t 's to obtain an expression for s_{t+1} (this is known as a proof by induction). Let's work out the case of s_3 in order to make this clear:

$$\begin{aligned} (1 + s_2)^2 (1 + f_{2,3})^1 &= (1 + s_3)^3 \\ \implies (1 + f_0)(1 + f_1)(1 + f_2) &= (1 + s_3)^3 \\ \implies s_3 &= \sqrt[3]{(1 + f_0)(1 + f_1)(1 + f_2)} - 1 \end{aligned}$$

(4) We know that the price of the 4-year zero-coupon bond is given by:

$$\begin{aligned}
 P &= \frac{\$10,000}{(1 + s_4)^4} \\
 &= \frac{\$10,000}{(1 + f_0)(1 + f_1)(1 + f_3)(1 + f_4)} \\
 &= \frac{\$10,000}{(1 + 5.26\%)(1 + 6.75\%)(1 + 7.23\%)(1 + 7.78\%)} \\
 &= \$7,700
 \end{aligned}$$

As we should expect, this is the same price we would get if we price the 4-year \$10,000 zero-coupon bond using discount factors.

Problem 9. (10 points) Suppose a Central Bank wants to achieve a steady inflation rate. Now suppose that it starts to see nominal interest rates rise in the financial market. How should it respond to this increase? Should it increase or decrease the money supply?

Solution. It depends on which components of the nominal interest rate are increasing and what the Fed's target for expected inflation is. In the simplest formulation, the nominal interest rate equals the real rate plus expected inflation. If the real rate is rising then the Federal Reserve will likely choose not to act since its ability to affect real rates of return, especially longer-term real rates, is transitory and limited. If expected inflation is increasing then, depending upon what the Fed's inflation target is, it may choose to decrease the money supply.

Problem 10. (10 points) Investors wonder how the Fed will react. The U.S. inflation rate is expected to be 1%, and the Congressional Budget Office estimates that current real GDP is 5% below potential GDP.

1. Use the Taylor rule to predict the federal funds target rate. Assume the Fed's inflation target is 2%, the long-run real interest rate is 2%, and the sensitivity parameters (or coefficients) for deviations of inflation from its target and for deviations of output from its potential are 1.5 and -0.5 respectively.
2. Why can't the Fed implement this policy target? Be specific.

Solution. (1) The Taylor rule is given by:

$$\begin{aligned}
 \text{Nominal Interest Rate} &= \text{Equilibrium nominal rate} + 1.5 \times (\text{Inflation} - \text{Inflation target}) - 0.5 \times (\text{Potential GDP} - \text{GDP}) \\
 &= 2\% + 1\% + 1.5 \times (1.5\% - 2\%) - 0.5 \times (5\%) \\
 &= -0.25\%
 \end{aligned}$$

(2) This is because of the “zero lower bound” on interest rates. Typically, the Federal reserve has been against setting the Fed Funds rate to be negative because they are concerned about the potential disruptions this could create for money market mutual funds, Treasury auctions, and the informational content of the Fed Funds rate.